

DS	Session	Tree 1	Question Information	Level 1	Challenge 65
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Question description

Given an array of n integers, your task is to process q queries of the form: what is the minimum value in range $[a,b]$?

Constraints

- $1 \leq n, q \leq 2 \cdot 10^5$
- $1 \leq x_i \leq 10^9$
- $1 \leq a \leq b \leq n$

Input

The first input line has two integers n and q : the number of values and queries.

The second line has n integers $x_1, x_2, \dots, x_n$: the array values.

Finally, there are q lines describing the queries. Each line has two integers a and b : what is the minimum value in range $[a,b]$?

Output

Print the result of each query.

Logical Test Cases

Test Case 1

INPUT (STDIN)

```
8 4
3 2 4 5 1 1 5 3
2 4
5 6
1 8
3 3
```

EXPECTED OUTPUT

```
2
1
1
4
```

Test Case 2

INPUT (STDIN)

```
10 4
13 12 14 51 11 1 5 3 4 6
3 4
3 6
4 8
1 3
```

EXPECTED OUTPUT

```
14
1
1
12
```

▼ Mandatory Test Cases

Test Case 1

KEYWORD

```
void build(int *aa,int k,int l,int r)
```

Test Case 2

KEYWORD

```
int query(int k,int l,int r,int ql,int qr)
```

▼ Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

3

Test Case 2

TOKEN COUNT

390

Test Case 3

NLOC

40

```
Tree15.c +
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 #define MAXN 100005
5 #define INF 2147483647
6
7 int tree[4 * MAXN];
8
9 void build(int *aa, int k, int l, int r) {
10     if (l == r) {
11         tree[k] = aa[l];
12         return;
13     }
14     int mid = (l + r) / 2;
15     build(aa, k * 2, l, mid);
16     build(aa, k * 2 + 1, mid + 1, r);
17     tree[k] = (tree[k * 2] < tree[k * 2 + 1]) ? tree[k * 2] : tree[k * 2 + 1];
18 }
19
20 int query(int k, int l, int r, int ql, int qr) {
21     if (qr < l || r < ql) return INF;
22     if (ql <= l && r <= qr) return tree[k];
23     int mid = (l + r) / 2;
24     int a = query(k * 2, l, mid, ql, qr);
25     int b = query(k * 2 + 1, mid + 1, r, ql, qr);
```

```
26     return (a < b) ? a : b;
27}
28
29int main() {
30     int n, q;
31     if (scanf("%d %d", &n, &q) != 2) return 0;
32     int *aa = (int*)malloc(n * sizeof(int));
33     for (int i = 1; i <= n; i++) scanf("%d", &aa[i]);
34     build(aa, 1, 1, n);
35     for (int i = 0; i < q; i++) {
36         int a, b;
```

```
37     if (scanf("%d %d", &a, &b) == 2) {
38         if (a > b) {
39             int t = a;
40             a = b;
41             b = t;
42         }
43         printf("%d\n", query(1, 1, n, a, b));
44     }
45 }
46 return 0;
47 }
```

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t1.txt

8 4

3 2 4 5 1 1 5 3

2 4

5 6

1 8

3 3

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Tree15.exe

2

1

1

4

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t2.txt

10 4

13 12 14 51 11 1 5 3 4 6

3 4

3 6

4 8

1 3

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Tree15.exe

14

1

1

12

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>